

Program overview

08-Dec-2022 11:17

Year 2022/2023
Organization Applied Sciences
Education Bachelor Molecular Science and Technology

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Bachelor MST 2022							
1e jaar BSc MST 2022	1st year MST 2022						
4051ALACHY	General and Inorganic Chemistry	6					
4051CALC1Y	Calculus 1	6					
4051CALC2Y	Calculus 2	3					
4051CHAN3Y	Methods of Chemical Analysis	3					
4051CHTHEY	Chemical Thermodynamics (CTD)	6					
4051INCHE	Inorganic Chemistry	3					
4051IPTECY	Introduction to Process Technology	6					
4051LEON1Y	Undergraduate Research Project 1 (LO1)	6					
4051OCSTRY	Organic Chemistry and Structural Analysis	9					
4051PRBVAY	Basic Laboratory Skills Course (PBV)	6					
4051QCHFYY	Quantum Chemistry and Physics	6					
2e jaar BSc MST 2022	2nd year BSc MST 2022						
Kernprogramma BSc MST 2e jaar 2022							
4052BIOC6Y	Biochemistry	6					
4052ENRV6	Energy, Recycling and Safety	6					
4052FYSCY	Physical Chemistry and Kinetics (FCK)	6					
4052KATALY	Catalysis (KAT)	3					
4052LEON2	Research Project 2	6					
4052NUMT3	Numerical Techniques	3					
4052STEVMY	Structure and Properties of Materials	6					
Major Materialen BSc MST 2e jaar 2022							
4052CHFVS	Chemistry and Physics of Solid Materials	6					
4052FYSTRY	Physical Transport Phenomena (FTV)	6					
4052LADIFY	Linear Algebra and Differential Equations	6					
4052THECHY	Theoretical Chemistry 1	6					
Major Synthese BSc MST 2e jaar 2022							
4052BMOCHY	Biomolecular Chemistry (BMC)	6					
4052FYSC6	Physiological Chemistry	6					
4052ORGC2Y	Organic Chemistry 2 (OC 2)	6					
4052THECHY	Theoretical Chemistry 1	6					
Major Technologie BSc MST 2e jaar 2022							
4052CHBIOY	Chemical Biotechnology (CBT)	6					
4052FYSTRY	Physical Transport Phenomena (FTV)	6					
4052LADIFY	Linear Algebra and Differential Equations	6					
4052SCHTEY	Separation Technology	6					
3e jaar BSc MST 2022	3rd year BSc MST 2022						
Kernprogramma BSc MST 3e jaar 2022							
4052LEON3	Chemical Product Design	6					
4052STAMEY	Statistical Methods (STM)	3					
Major Synthese en Materialen BSc MST 3e jaar 2022							
4052LEON4	Bachelor Thesis Project	15					
4052STTS6	Statistical Thermodynamics / Spectroscopy	6					
Major Technologie BSc MST 3e jaar 2022							
4052CHREKY	Chemical Reactor Engineering	6					
4052LEON4	Bachelor Thesis Project	15					

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Bachelor MST 2022	
In association with the Faculty of	TNW
Administration by the Faculty of	TNW

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

1e jaar BSc MST 2022

4051ALACHY	General and Inorganic Chemistry	6
Responsible Instructor	Dr. E.M. Kelder	
Instructor	Dr. A.P.A. Janssen	
Contact Hours / Week x/x/x/x	16/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Expected prior knowledge	VWO Chemistry	
Course Contents	The course offers a general view of the inorganic and physical fundamentals of chemistry. Part 1: A description of matter at the microscopic level, with an emphasis on atomic and molecular structure. Part 2: The structure of solids, description of gasses and liquids/solutions, phase transitions. Introduction to the chemical conversion of compounds, with an emphasis on acid-base and electrochemical reactions; the thermodynamic and kinetic description of reactions and the origin of equilibria. Emphasis is repeatedly focused on the Lithium-ion battery, the fuel cell and the production and storage of hydrogen.	
Study Goals	Upon completion of the course, the student will recall the molecular structure of materials and is able to explain the relation between atomic/molecular structure and macroscopic properties. The student is familiar with the elementary principles of thermodynamics and reaction kinetics and is able to apply these to chemical reactions and equilibria, in particular with regard to acid-base reactions and electrochemistry. Together with fellow students, the student knows how to study an inorganic subject in depth and present the essentials in an oral presentation. The student can use Microsoft Excel for data storage, for the calculation of inorganic and physic-chemical parameters and for the reproduction of simple functions.	
Education Method	Classroom lectures, tutorials, self and group study, demonstrations	
Literature and Study Materials	Will be handed out	
Assessment	Two partial exams and a daily progress test. A resit over the entire material. (First exam: 35%, second exam: 60%, daily quizzes: 5%).	
Enrolment / Application	All students who start the Molecular Science and Technology program for the first time are registered by the program for all courses and exams of the first semester. The following applies to other students: to participate in this course, registration in uSis is required, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE NEW: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

4051CALC1Y	Calculus 1	6
Responsible Instructor	Dr. C.O. Smet	
Contact Hours / Week x/x/x/x	16/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Expected prior knowledge	VWO Mathematics B	
Course Contents	Calculus course with the subjects Complex numbers, Limits and differentiation, Integrate, Differential equations, Series, Vectors lines and planes.	
Study Goals	Complex numbers: - The student can perform arithmetic operations with complex numbers, determine the modulus and argument of a complex number, write a complex number as e-power and make a sketch of the complex number in the complex plane. - The student can solve quadratic equations and determine the nth roots of a complex number. Limits and differentiation: - The student knows the concepts of limit, continuity, differentiability and differential. - The student can determine limits by clamping, by distributing the main element, with the rule of the l'Hospital, and with Taylor polynomials. - The student can differentiate implicitly, and determine tangents to curves and linearisations of functions. Integration: - The student can calculate simple, indefinite and specific (single) integrals, if necessary by using: substitution, fractional splits and partial integration. - The student knows the (properties of) the inverse functions arcsin, arccos and arctan. - The student recognizes improper integrals and can determine whether they are convergent or divergent. Differential equations: - The student can prepare a differential equation for some physical situations. - The student can solve 1st-order differential equations by filling in an e-power, using separation of variables or by means of an integrating factor. - The student can approach the solution with the Euler method. - The student can solve both homogeneous and simple inhomogeneous, linear 2nd-order differential equations with constant coefficients. - The student can describe solutions of the form $A \cos(\omega t) + B \sin(\omega t)$ in the form $C \cos(\omega t - \alpha)$. Series: - The student can use the sum notation. The student knows the definitions of convergent and divergent rows and series. The student can determine the limit of a convergent recursively defined row. - The student recognizes geometric series and p series and knows when they are convergent or divergent. The student can determine the sum of a convergent geometric series. - The student can demonstrate divergence of series with the nth-term test and apply the integrated test to demonstrate convergence or divergence. - The student can approach a function with a Taylor polynomial and estimate the error at a certain interval. Vectors, lines and surfaces: - The student can add vectors and scalarly multiply. The student can calculate products, lengths, distances, angles and determinants internally and externally. The student can calculate surfaces of triangles and parallelograms and volumes of tetrahedra and parallelepiped. - The student can draw up an equation for a sphere with a center point and a radius, and can conversely (possibly split off with squares) determine the center point and radius from an equation for a sphere - The student can determine parameter representations of lines and planes, and with the aid of a normal vector determine an equation for a plane.	
Education Method	There are 8 lecture-instruction hours per week. In addition, there is supervised self-study 8 hours per week.	
Literature and Study Materials	Calculus: Early Transcendentals, 7th edition van Edwards & Penny ISBN:9781292022178	
Assessment	There are four tests, the results of which respectively count 10%, 30%, 30% and 30% in the final mark. The tests cannot be made up or retaken. Afterwards there is a resit regarding the complete content of the course.	
Enrolment / Application	All students who start the Molecular Science and Technology program for the first time are registered by the program for all courses and exams of the first semester. The following applies to other students: to participate in this course, registration in uSis is required, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE NEW: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

4051CALC2Y	Calculus 2	3
Responsible Instructor	Dr.ir. V. van Steijn	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Expected prior knowledge	4051CALC1Y: Calculus 1	
Course Contents	Functions of more variables, multiple integrals, linear algebra	
Study Goals	From functions of more variables, the student can determine partial derivatives, interfaces, extreme values, directional derivatives and gradients. The student can use these quantities to approach and optimize. The student can integrate functions over two- and three-dimensional areas, also with the help of polar, cylinder and sphere coordinates. The student can solve systems of linear equations and give the solutions in a parameter representation. The student can multiply and invert matrices. The student can determine the determinant and values and vectors of 2x2 matrices.	
Education Method	Lecture and hours for self-study in which students can make assignments themselves under the supervision of assistants.	
Literature and Study Materials	Calculus: Early Transcendentals, 7th edition van Edwards & Penny ISBN:9781292022178	
Assessment	The exam consists of two partial tests, the first halfway through, and the second at the end of the lecture. Both have the same weight. If the average of both tests is insufficient, the student must do a resit for the entire course in the week after the second period.	
Enrolment / Application	All students who start the Molecular Science and Technology program for the first time are registered by the program for all courses and exams of the first semester. The following applies to other students: to participate in this course, registration in uSis is required, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE NEW: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

4051CHAN3Y	Methods of Chemical Analysis	3
Responsible Instructor	Dr. B.I. Florea	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	Successful completion of the Basic Skills Practicum (including error analysis) and the General and Inorganic Chemistry course contribute to a better understanding of the material.	
Course Contents	Experimental error, statistics, quality control, introduction to analytical separation techniques, high performance liquid chromatography, other chromatographic methods and electrophoresis, spectrophotometry, mass spectroscopy	
Study Goals	Upon completion of the course, the student will have obtained insight to considering experimental errors and the basic principles of chromatography and spectroscopic methods	
Education Method	Lectures and tutorials.	
Literature and Study Materials	The lecture uses the book of D.C.Harris: Quantitative Chemical Analysis, D.C. Harris, 8th edition, ISBN-10: 1-4292-3989-1	
Assessment	Written exam and a retake.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

4051CHTHEY	Chemical Thermodynamics (CTD)	6
Responsible Instructor	Dr. T.J. Savenije	
Instructor	Dr. B. Bera	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	4051CALC1Y: Calculus 1	
Course Contents	Fundamental concepts Heat, work and internal energy The importance of state functions Thermochemistry Entropy and the second and third law of thermodynamics Chemical Equilibrium The properties of real gasses Phase diagrams and the relative stability of solids, liquids and gases Ideal and real solutions	
Study Goals	After studying the course, the student must: a. have knowledge of the basic concepts and laws of thermodynamics; b. be able to qualitatively predict the behavior of simple thermodynamic systems on the basis of thermodynamic laws; c. being able to do simple processes and systems based on the learned quantitative calculations You should be able to answer questions such as: What are the laws governing the behavior of an (ideal) gas? What is the specific heat? Why is it impossible to convert heat completely into work? What is the energy available for work? What is an adiabatic process? What is a reversible process? What is a Carnot cycle? How to increase the efficiency of cycles. Calculate the entropy, enthalpy and Gibbs energy of a chemical reaction and their dependence on temperature and pressure. Determine the equilibrium constant of a reaction and its dependence on temperature and pressure. What is the difference between real and ideal gasses? How do you calculate the osmotic pressure?	
Education Method	Lecture and guided working classes.	
Literature and Study Materials	Physical Chemistry Thermodynamics, Statistical Thermodynamics and kinetics, Chapter 1-9, 4rd Ed, Engel and Reid, Pearson Publishers, including the electronic set of tutorials/question 'Mastering Chemistry'.	
Assessment	The education consists of lectures and computer-assisted seminars. At the end of this period a written exam is taken on the material. An interim test is also provided. The grade is determined by the results of the seminar (15%), the interim test (20%), and the final test (65%). The resit exam covers the entire material. The results of the final test and the interim test do not count in this case. The (re-) exam is a written exam.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	More information via Brightspace (Leiden). It is necessary to have a login code for Mastering Chemistry. The purchase of a new book is therefore necessary.	
Location	Delft	

4051INCHE	Inorganic Chemistry	3
Responsible Instructor	Prof.dr. E. Bouwman	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Course Contents	Concepts from coordination chemistry are explained on the basis of important applications of metals in nature, in medicine, in materials or in homogeneous catalysis. These terms include: nature of coordination binding; coordination number, symmetry, geometry and isomerism of complexes; 18-electron rule; stability and reactivity of complexes; spin states and magnetism; oxidation states and redox chemistry, ligand field breakdowns and ligand field stabilization energy.	
Study Goals	After passing the exam: - the student is able to explain and apply the different aspects of coordination chemistry (including isomerism, jargon). - the student can reason the ligand field breakdown of d-orbitals in different geometries. - the student can make a judgment about the stability and reactivity of a random coordination connection. - the student is familiar with important applications of coordination chemistry in, for example, medicine and catalysis.	
Education Method	Lectures and tutorials	
Literature and Study Materials	Inorganic Chemistry 7th Ed., M. Weller, T. Overton, J. Rourke, and F. Armstrong, Oxford University Press, 2014. ISBN: 978-0-19-876812-8. Handouts of the lectures, exercises used in the tutorials.	
Assessment	Written exam and a retake.	
Permitted Materials during Tests	None. A periodic table will be provided at the exam.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

4051IPTECY	Introduction to Process Technology	6
Responsible Instructor	Ir. R.M. de Kruijff	
Instructor	Dr.ir. V. van Steijn	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Mathematics ("Calculus 2 (CA2)") - solving systems of linear equations, Thermodynamics ("Chemische Thermodynamica (CTD)") - single phase: ideal gas law, equations of state - gas-liquid equilibrium (Raoult, Henry, Txy diagram, lever rule) - Gibbs phase rule - first law of thermodynamics (specific internal energy, specific enthalpy, heat, work, enthalpy of formation, enthalpy of reaction, standard conditions, references) Physics ("Quantumchemie en Fysica (QCF)") - energy balance: kinetic energy, potential energy	
Course Contents	Solving elementary problems in process technology through application of mass balances, energy balances and equilibrium relationships.	
Study Goals	By the end of the course, students should be able to: 1. Recognize engineering roles in society. 2. Recognize the nomenclature of chemical engineering. Know systems of units and dimensions. Define and relate process variables. Learn basic unit operations of chemical processes and describe how each works qualitatively. 3 (!!!). Use a systematic approach to solve chemical engineering problems. Identify variables, draw and label a process flow chart from a word description. Perform and use degrees of freedom analysis. Formulate mathematical expressions that represent word problems. 4. Work effectively in teams. Recognize the skills needed to function in a team. Write a clearly structured report.	
Education Method	Lectures, working classes, question hours, obligatory excursion to a chemical plant (including an assignment and written report).	
Literature and Study Materials	Felder's Elementary Principles of Chemical Processes, 4th Edition, Global Edition Powerpoint presentations of lectures and exercises. Recorded past lectures and explanatory video material.	
Assessment	Two written tests. Test are cumulative, having a weight of 0.20 and 0.80 respectively. In case the cumulative grade of the tests is below the passing mark, a retake exam can be done. Besides a passing mark based on the tests, the report on the obligatory group assignment related to the excursion should be of satisfactory quality. In case the report is of excellent quality a maximum of half a bonus point can be obtained, which will be awarded provided that the student has a passing cumulative grade for the tests. NOTE: Depending on the COVID-19 measures the examination method can be adapted during the academic year.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

4051LEON1Y	Undergraduate Research Project 1 (LO1)	6
Responsible Instructor	Dr. W.F. Jager	
Course Coordinator	M.P. Swarte	
Contact Hours / Week x/x/x/x	0/0/0/16	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Practical basic skills (PBV)	
Course Contents	Students carry out experiments in small groups as part of ongoing research at one of the research groups in Leiden or Delft.	
Study Goals	1. Literature search. Knowledge of relevant literature and data sources 2. Getting ready for practical experimentation: The capability to prepare experiments (define the goals, describe the experimental details); Environment and safety: the ability to work safely with chemicals, taking into account their physical and chemical properties, with special regard for the possible specific dangers that are associated with these compounds; Proficiency in making a risk analysis of the use of chemicals. 3. The gathering and compilation of experimental data: Dexterity in carrying out standard laboratory experiments, as well as in the use of instrumentation for synthetic and analytical tasks; Dexterity in following the progress of the experiment, by collecting data from observation and measurements and reporting them correctly (as e.g. in a lab journal). 4. The analysis and processing of experimental data: The evaluation and interpretation of observations and data collected in the laboratory and the ability to relate these to an appropriate theory; Ethics: perception of ones professional and ethical responsibility. 5. Employment attitude and teamwork: Dedication, motivation, enthusiasm, creativity; Capability to work in a (multicultural and multidisciplinary) team; Active participation, effective cooperation, effectiveness in discussions, social contacts. 6. Reporting: Ability to report, both orally and in writing, with proper regard for the requirements demanded by scientific communication and responsiveness; Proficiency in effective communication, in the Dutch as well as in the English language, thereby using modern ways and means of presentation.	
Education Method	Practical work in a research group. Attending work meetings. Self study and group work. A compulsory Oral Presenting course is part of Learning Research 1.	
Literature and Study Materials	Literature is provided by each research group.	
Prerequisites	Participation in Learning Research 1 (LO1) is only possible after successful completion of the Practicum Basic Skills (PBV) and when the access requirements that apply to the building and the laboratories where the practical is performed.	
Assessment	The assessment takes place on the basis of the learning objectives set. This means that you must show whether you have actually mastered the abovementioned skills and knowledge. This can (among other things) be based on an excellently kept journal, experimental work, good time planning, working in with others in a team, oral and written reports and based on the impression of the supervisors during the project. Participation in Leren onderzoeken 1 symposium with an presentation is mandatory.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. PLEASE NOTE This is a course with a practical exercise: registration closes 28 days before the start of the course;	

4051OCSTRY	Organic Chemistry and Structural Analysis	9
Responsible Instructor	Dr. S. van der Vorm	
Instructor	Dr. D.V. Filippov	
Contact Hours / Week x/x/x/x	0/16/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Expected prior knowledge	4051ALACHY: Algemene en Anorganische Chemie	
Course Contents	Functional groups, hybridization and orbitals. Delocalization and conjugation. Inductive effects and steric hindrance. Acids and bases and their qualitative strength. Reactions of aldehydes, ketones, carboxylic acids and derivatives and the corresponding mechanisms. Aromaticity. Thermodynamics and kinetics of reactions, mechanisms. Introduction to spectroscopic methods: UV-VIS, 13 C-NMR and 1 H-NMR, chemical shift, spin-spin coupling, IR spectroscopy Stereochemistry and chirality. Enantiomers and diastereomers, racemic mixtures. Conformation analysis, Newman projection, 1.3 diaxial interaction. Substitution reactions of aliphatic compounds and associated mechanisms. Nucleophilicity and leaving group. Electron-sucking and driving substituents Heteronuclear coupling, strong coupling in 1 H-NMR, Karplus relation, "high order" phenomena in NMR spectra, structure clarification. Elimination of aliphatic compounds. Stereo selectivity and stereo specificity. Electrophilic addition, the chemistry of enols and enolates. Electrophilic and nucleophilic aromatic substitution. Conjugated addition. The corresponding response mechanisms. NOE effects in NMR, two-dimensional NMR (1 H-1 H and 13 C-1 H correlated spectra). Dynamic effects in NMR.	
Study Goals	Learning objective 1 Ready knowledge and application of functional groups, the orbitals and their reactivity, fundamental concepts such as conjugation, aromaticity. Learning objective 2 Insight into the mechanism of the most important reactions that the functional groups can undergo in terms of electron movement and orbitals, acid and base strength, configuration and conformation Learning objective 3 Skills in interpreting spectroscopic data of organic compounds. Insight into the relationship between the structure of organic compounds and their spectra	
Education Method	lecture / tutorials	
Literature and Study Materials	Organic Chemistry, second edition, J.Clayden, N Greeves and S.Warren ISBN: 9780199270293	
Assessment	The final grade is determined by the following weighting: test-1 once, test-2 twice test-3 three times. Test 1 is a compilation of small preparation-tests taken during the lectures.	
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Special Information	send email to: s.van.der.vorm@chem.leidenuniv.nl	
Continuing Courses	Organische Chemie 2 (Leiden)	
Location	Leiden	

4051PRBVAY	Basic Laboratory Skills Course (PBV)	6
Responsible Instructor	Dr. R. Eelkema	
Responsible Instructor	Dr. R.J.B.H.N. van den Berg	
Contact Hours / Week x/x/x/x	8/8/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	Dutch	
Required for	LO1	
Expected prior knowledge	VWO chemistry and a passed safety test	
Course Contents	The training in practical skills in the areas of synthesis (organic, inorganic and biochemical) and technology (chemical, physical and biotechnology). Safety instruction. Report writing. ICT (Excel, Python). Data processing. Error analysis and propagation.	
Study Goals	This course teaches first year students basic practical skills. Upon completion of the course, the student has acquired sufficient skills to be able to start practical experimentation in one of the research groups (see: Research Oriented Learning).	
Education Method	Practical Course	
Literature and Study Materials	Course Manual and web support	
Assessment	measurement reports, observed practical skills	
Enrolment / Application	All students who start the Molecular Science and Technology program for the first time are registered by the program for all courses and exams of the first semester. The following applies to other students: to participate in this course, registration in uSis is required, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE NEW: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Special Information	With the responsible teachers Richard van den Berg and Rienk Eelkema.	
Location	1st period: Leiden or Delft 2nd period: Delft or Leiden	

4051QCHFYY	Quantum Chemistry and Physics	6
Responsible Instructor	Prof.dr. L.D.A. Siebbeles	
Instructor	J. Meyer	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	4051CALC1Y: Calculus 1	
Course Contents	In this course, in the first part, classical physics as discussed in high school is repeated and deepened. This is followed by an introduction to quantum mechanics with specific attention to quantum effects that are important in atoms and molecules. Topics that are covered are: - Classic mechanics according to Newton - Oscillations and rotations - Running and standing waves - Electrostatics - Limitations of classical mechanics - Basic concepts in quantum mechanics - Schrodinger comparison solutions - Particle in a box - Quantum description of harmonic oscillator and rigid rotator - Electron structure of atoms and simple molecules	
Study Goals	- The student is familiar with the basics of classical physics, consisting of Newton mechanics, standing and running waves, oscillations and rotations and electrostatics - The student knows the limitations of classical physics and can argue in which cases a quantum mechanical description is needed - The student knows the basic principles of quantum mechanics and can apply these to simple systems - The student has a basic understanding of the electron structure of atoms and molecules	
Education Method	Lectures are alternated with tutorials. Use is made of the online learning environments Mastering Physics and Mastering Chemistry.	
Literature and Study Materials	Physics for Scientists and Engineers , Auteur: Randall D. Knight, 4e editie, ISBN-10 1-292-15742-9 Quantum Chemistry and Spectroscopy, Auteur: Thomas Engel, Pearson New International Edition, 3rd Edition	
Assessment	The course is tested by means of an interim test (30%) and a written exam (70%), each of which partially determines the final grade. The resit is over the entire material.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	E-mails from teacher J. Meyer please use this address: j.meyer@chem.leidenuniv.nl	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

2e jaar BSc MST 2022

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Kernprogramma BSc MST 2e jaar 2022

4052BIOC6Y	Biochemistry	6
Responsible Instructor	Dr. R.R.C.L. Olsthoorn	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course Contents	Amino acids, peptides, proteins, (primary quaternary structure), enzymes, enzyme kinetics, enzyme regulation, lipids, membranes, membrane proteins, muscle proteins, general principles of metabolism and important metabolic routes. DNA and RNA structure, DNA replication, DNA transcription, RNA splicing, translation	
Study Goals	Part I 1. Know: Structure and properties of the 20 main amino acids Protein structure: Secondary and tertiary structural elements such as alpha helix and beta sheet. The general structure of some fibrous proteins and haemoglobin 2. Can: Ligand binding and Enzyme catalysis: calculating K_d, know the Michaelis-Menten equation and how to apply it, effects of inhibitors on Michaelis-Menten. 3. Understanding of Membrane proteins, transmembrane domains, Potassium channels, membrane fusion and the mechanism of chymotrypsin 4. Recognize: Structure and properties of: Sugars: mono-, di- and polysaccharides (starch, glycogen and cellulose) Lipids: triacylglycerol, glycerophospholipids and sphingolipids 5. Understanding of metabolism: Seminars provide insight into the functioning of important metabolic pathways, such as glycolysis, gluconeogenesis, citric acid cycle, oxidative phosphorylation, pentose-phosphate pathway, photosynthesis. Part II: 6. Know: Structure and properties of building blocks of DNA and RNA (nucleotides) DNA and RNA secondary and tertiary structural elements, such as the double helix. Thermodynamic and chemical stability of DNA and RNA. Structure of chromosomes and ribosomes 7. Know how the following processes work: DNA replication and repair. DNA sequencing Transcription in prokaryotes and eukaryotes, and reverse transcription RNA processing: splicing of mRNA, ribosomal RNAs, tRNA and microRNAs Translation of mRNA into eukaryotes and prokaryotes Translation inhibition by antibiotics and microRNAs	
Education Method	Lectures and tutorials	
Literature and Study Materials	Principles of Biochemistry, Lehninger, 8e editie. editors Nelson & Cox. ISBN-10: 978-1-319-38149-3.	
Assessment	2 written partial exams (50% / 50%). The resit is over the entire course material. For the 2nd partial exam, you can also earn 10 bonus points by means of: a short multiple-choice test that will be taken on-campus during one of the lectures. These bonus points can only be used if your score for the 2nd partial exam is higher than 55 points, in other words you cannot use them to make a pass from an unsatisfactory grade.	
Enrolment / Application	PLEASE NOTE: Due to the measures surrounding the Coronavirus, the method of testing may change during the academic year. Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

4052ENRV6	Energy, Recycling and Safety	6
Responsible Instructor	Prof.dr. S.J. Picken	
Instructor	Prof.dr. J.J.C. Geerlings	
Instructor	Prof.dr.ir. G.L.L.M.E. Reniers	
Instructor	Drs. W.J. Blokzijl	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Course Contents	The energy, recycling and safety course focuses on the safe and sustainable use of energy and raw materials. It consists of a theoretical framework and teamwork (5-6 students per team). Topics to be discussed are: Energy: - Relationship between availability of energy and prosperity / well-being - World primary energy consumption, fossil energy - Population growth; carrying capacity of the earth - Earth energy balance, greenhouse effect, climate change - The "Energy transition". Learning curves renewable energy technologies (PV, Wind etc), energy storage, material intensity renewable energy - "Renewable Energy for Industry (IEA)" Recycling: - Main classes of materials - Plastics and recycling (application dependent, types of plastics) - Biobased plastics (what can you do with them, what was the intention) - The various types of reuse "downcycling", "recycling" & "upcycling" (lower, same, higher value after reuse) - Aspects related to total fraction of fossil => polymers (approx. 5% max.10%, CO2 footprint) - Design of products versus reusability (e.g. PET, nylon = OK, metallized laminate packaging film = NOT OK) - Pro's and con's of polymers/plastics, replacement/alternatives, dilemmas, context in society Safety: -Introduction to risk and safety, conceptual framing and typing -Risk Management: Safety history, legislation, risk management models, location-related risk calculation and group risk -Process safety, safety in the process industry: what is specific to it, what are new trends? What are major accidents that have been compromised and the lessons learned? -Human and Organisational Factors of Safety (HOFS): Safety Culture (climate) and framing in performance management -Risk Analysis Techniques: discussion, classification Academic skills: - written reporting (structure, structure and argumentation - report) - teamwork	
Study Goals	1. Knowledge a. The student is familiar with the material covered in the themes of energy, recycling and safety and can answer questions about this b. The student is familiar with the way in which a report/article/publication must be set up. 2. Insight and understanding a. The student is able to propose solutions for current issues b. The student is able to develop new ideas c. The student is able to model certain aspects. 3. Apply the substance a. The student is able to apply the subject matter of the course to issues from current practice. b. The student is able to make the correct assumptions for modeling the practical situation. c. The student is able to work out a topic on an academic level in a team and to present this in a good (scientific or popular) article/publication/report.	
Education Method	Lectures and teamwork	
Books	1. Cedric Philibert, Renewable Energy for Industry, International Energy Agency, Insight Series 2017 (pdf on internet) 2. A.K. van der Vegt, L.E. Govaert, Polymeren van keten tot kunststof, ISBN 9789071301483, VSSD (recommended) 3. Thierry Meyer, Genserik Reniers, Engineering Risk Management, 2nd Edition, ISBN 978-3-11-041803-3	
Assessment	- The final grade is composed of two parts: 1. Written exam (ERT): 50% of the final mark; The result must be at least 6.0. 2. Report / article (SV) of the team: 50% of the final mark; The result must be at least 6.0 - Final grade is calculated as follows: Final grade = 0.5*ERT + 0.5*SV - Final grade on a half or whole - Rounding: 5.5 is rounded to a 6.0 - All components can be retaken: - 1. Exam via resit - 2. Report / Article only supplementary up to a maximum of 6.0. - For students who have passed the writing assignment on safety in the previous academic year: report to the lecturer responsible at the start of the course to be exempted from the writing assignment and to complete the recycling & energy components this year by means of the (resit) exam. Please note: this exemption only applies if the writing assignment was successfully completed in the previous year, not for previous years.	
Enrolment / Application	PLEASE NOTE: Do to possible measures related to the Coronavirus the mode of examination might have to be different. - Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate. - The report / article must be submitted on a date that will be announced (digitally on Brightspace before 5 p.m.)	

4052FYSCKY	Physical Chemistry and Kinetics (FCK)	6
Responsible Instructor	L. Juurlink	
Instructor	Dr. R. Mom	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Expected prior knowledge	General and Inorganic Chemistry, Chemical Thermodynamics, Introduction to Process Technology, Quantum Chemistry and Physics, Calculus 1 and 2	
Course Contents	Component Physical Chemistry: Ideal and real solutions; electrolyte solutions (thermodynamics, activities); electrochemistry (Nernst equation, different kinds of cells, fuel cells, batteries); transport (diffusion, migration); electrode kinetics (Butler-Volmer, kinetics and transport) Component Kinetics: Kinetic gas theory; elementary chemical kinetics: reaction rate equations based on elementary reactions, sequential and parallel reaction steps, (perturbation of) equilibria and temperature dependence; kinetics of complex reactions: pre-equilibrium, Lindemann mechanism, homogeneous, heterogeneous and enzymatic catalysis, radical reactions and explosions; the build up of surface potentials, elementary molecular reaction dynamics. Complex kinetic processes are being actively studied by applying the requisite mathematical models, with the aid of, e.g.: Excel and / or Maple. To this effect, each student receives homework exercises.	
Study Goals	Component Physical Chemistry: Upon completion of the course, the student: - knows the properties of an ideal solution and is able to formulate deviations of ideal behaviour; - knows the thermodynamic behaviour of electrolyte solutions and can apply the Debye-Hueckel theory; - can apply the Nernst equation and is able to differentiate between different kinds of electrochemical cells; - understands the molecular basis of transport, diffusion and migration; - is knowledgeable about the elementary aspects of the kinetics of electrode reactions; - is able to apply simple mathematical models of chemical reactions at electrochemical and catalytic surfaces, in combination with mass transport. Component Kinetics Upon completion of the course, the student: - is able to correctly assimilate, with MS Excel, experimental data in order to determine kinetic parameters, such as the order of a reaction, reaction rate constants and activation barriers; - is able to correctly interpret crucial aspects of reaction mechanisms on the basis of kinetic parameters; - can produce a simple mathematical model of a reaction mechanism with Maple, calculate a solution and display this graphically; - can describe, in a semi-quantitative way, the build up of, and the movement on, a potential surface.	
Education Method	Lectures, tutorials, self-study, demonstrations	
Literature and Study Materials	1. Physical Chemistry, Engel & Reid, Pearson, 2006 ISBN: 0-8053-3842-X 2. Microsoft Excel 3. Maple 9 of 10 (beschikbaar op computers van de TU Delft en Univ. Leiden)	
Assessment	Written exam and retake.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

4052KATALY	Catalysis (KAT)	3
Responsible Instructor	M.T.M. Koper	
Instructor	Prof.dr. U. Hanefeld	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	Dutch	
Expected prior knowledge	4051ALACHY: Algemene en Anorganische Chemie 4051OCSTRY: Organische Chemie en Structuuranalyse	
Course Contents	The student is offered a kaleidoscopic view of catalysis. Similarities and disparities between different kinds of catalysis are illustrated with a number of reactions that can be catalysed in different ways. Examples are the reduction of nitrogen oxide, the oxidation of carbon monoxide, the hydrolysis of nitriles and the selective reduction of carbon-carbon double- and triple bonds. Emphasis is on the basic principles.	
Study Goals	Upon completion of the course, the student: - is knowledgeable about a number of catalytic processes, both homogeneous, heterogeneous, bio-catalytic and electrocatalytic; - has obtained insight in the most important similarities and dissimilarities, advantages and disadvantages and range of application of the four different areas of application of catalysis; - is skilled in the application of the principles of chemical reaction kinetics to catalytic processes and mechanisms; - has a clear understanding of the mode of action, the molecular mechanism, the preparation, the characterisation and the stability of a number of catalysts and catalytic processes in the four areas of catalysis; - knows about the possibilities to immobilize a catalyst; - is well aware of the important applications of catalysis in industry.	
Education Method	Lectures, tutorials and self-study	
Literature and Study Materials	Handouts and slides	
Assessment	Written exam and retake.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

4052LEON2	Research Project 2	6
Responsible Instructor	Dr. W.F. Jager	
Course Coordinator	M.P. Swarte	
Contact Hours / Week x/x/x/x	0/0/16/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch English	
Course Contents	Students independently carry out experiments in the field of a current research of a research group at the universities of Leiden or Delft.	
Study Goals	The study goals for Research Project 2 are part of the Leren Onderzoeken (Learning Research) learning line for practical research, and follow up on Basic Laboratory Skills Course and Research Project 1. a. The student can find relevant scientific articles and books independently and critically assess their content. b. The student can design and prepare an experiment independently. Takes proactive measurements to ensure safety. c. The student can independently conduct an experiment and document it properly using a lab journal/notebook. d. The student can critically analyse and interpret observations and data acquired in the lab and state how reliable these results are. e. The student shows a productive attitude (effort, motivation, enthusiasm) and shows that they can work in a multicultural and multidisciplinary team. The student actively seeks out the attendant to discuss results. f. The student can independently write a report (10 pages) and provide arguments and defences for their results.	
Education Method	Working as a young researcher in the laboratory at a research group.	
Prerequisites	Participation in Learning Research 2 (LO2) is only possible after successful completion of LO1, and when the entry requirements that apply to the building and the laboratories where the practical is performed have been met.	
Assessment	The review will be made on the basis of the learning objectives set ("Rubric "). This means that you have to show whether you have actually shown your mentioned skills and knowledge. This can be (a.o.) based on an excellently maintained lab journal, experimental work, good time planning, working in a team, verbal and written reports and on the basis of the impression of the supervisors during the project.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. PLEASE NOTE This is a course with a practical exercise: registration closes 28 days before the start of the first part of the course;	

4052NUMT3	Numerical Techniques	3
Responsible Instructor	Dr. F.C. Grozema	
Contact Hours / Week x/x/x/x	8/0/8/0	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	Dutch	
Course Contents	The numerical techniques course focuses on the use of numerical techniques for solving complex issues using a computer. The course has a focus on the use of such numerical techniques for solving problems in chemistry and chemical engineering. The following topics are dealt with: 1. Introduction Python; Introduction programming and algorithms; 2. Statistical analysis and plotting of data; 3. Curve fitting and regression; 4. Numerical integration and differentiation; 5. Solving (systems of) linear and nonlinear equations; 6. Numerically solving systems of differential equations; 7. Monte Carlo Methods.	
Study Goals	1. Knowledge A. The student is familiar with the basic concepts of (structured) programming. B. The student is familiar with the most common numerical methods used in solving chemical and "chemical engineering" issues. 2. Understanding and understanding A. The student is able to indicate how big the error is in the different numerical methods. B. The student is able to estimate the results of a numerical calculation by value. 3. Application and analysis of the subjects A. The student is able to apply the subjects of the course to issues in the field of chemistry and chemical engineering. B. The student is able to assess when applying a numerical technique makes sense.	
Education Method	Lectures and instruction where practical chemical technological problems are solved with the help of the numerical methods that are programmed in Python.	
Assessment	The final grade is determined by a (computer) exam where chemical/engineering problems need to be solved numerically by writing Python programs.	

4052STEVMY	Structure and Properties of Materials	6
Responsible Instructor	Prof.dr. J.H. van Esch	
Instructor	Dr. E. Mendes	
Contact Hours / Week x/x/x/x	8/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Expected prior knowledge	Basis of Physical Chemistry (B.Sc. level)	
Course Contents	"Soft Matter" includes a class of materials from everyday life that exhibit characteristics of both liquids and solids. Such materials are found in a large part of traditional chemical & food products as well as in high-tech applications ranging from nano-medicine to energy. This course offers an extensive overview of "soft materials", their properties, analysis methods and synthetic strategies and explores the relationship between their structure, dynamics and properties. Moreover, the course also illustrates how their properties can be used in traditional and high-tech chemical products. The course starts with the chemical synthesis, molecular forces and principles needed to describe the elementary theories that underlie the structure and dynamics of these materials. It then develops further into the description of their microscopic and macroscopic properties. We will then examine how these quantities can be determined experimentally. The course deals with systems such as polymers, surfactants, dispersions and liquid crystals and learns how the different types of molecular interactions can be used in (advanced) applications	
Study Goals	After successfully completing this course, students will be able to: - describe and discuss the properties of soft materials - calculate basic properties that characterize these materials - use elementary theories to describe the structure, dynamics and properties of soft materials - choose the appropriate characterization method for the evaluation of a certain characteristic - Understand and critically discuss basic concepts underlying the behavior of soft materials - to interpret quantitative experimental data - describe how molecular interactions and material structures can be used in an industrial application - propose molecular systems that are suitable for a specific application	
Education Method	The course consists of a series of lectures, accompanied by tutorials and self-study. In the lectures the most important content of the course is discussed with the help of material from textbooks. During self-study, students prepare for the lectures, rehearse the material from the lectures, and learn how to apply this material to problem solving. The solutions to the problems will be discussed during the tutorials.	
Literature and Study Materials	Introduction to Soft Matter: Synthetic and Biological Self-Assembling Materials, by Ian W. Hamley, Wiley	
Books	Introduction to Soft Matter: Synthetic and Biological Self-Assembling Materials, by Ian W. Hamley, Wiley	
Assessment	Written exam and a retake.	
Permitted Materials during Tests	Pen, paper, scientific calculator	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Major Materialen BSc MST 2e jaar 2022

4052CHFVS	Chemistry and Physics of Solid Materials	6
Responsible Instructor	Dr. I.M.N. Groot	
Responsible Instructor	Dr. T.J. Savenije	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Chemische thermodynamica, quantumchemie en fysica, calculus 1 en 2, anorganische chemie.	
Course Contents	In this course the fundamental chemical and physical properties of solids and materials will be discussed. We will start with how solids are built up by discussing the bonds between atoms. From there we continue with crystal structures, Miller indices, the reciprocal space and Brillouin zones. Gradually we will look more and more at the properties of materials that arise from their structure. basic chemical and physical properties of solids and materials. We will start with how solids are built up by discussing the bonds between atoms. From there we continue with crystal structures, Miller indices, the reciprocal space and Brillouin zones. Gradually we will look more and more at the properties of materials that arise from their structure. Essential prior knowledge: chemical thermodynamics, quantum chemistry and physics, calculus 1 and 2, inorganic chemistry. The following topics will be covered: - Theory of atoms and molecules, including the Bohr model and the quantum mechanical model, and the Schrödinger equation - Theory of solids, including connections in metals, tire structure, and electronic structure - Crystal structures, including X-ray diffraction and Miller indices - Transport properties of solids, including metals, insulators and semiconductors - Optical properties of solids - Diffusion in solids, including thermodynamics and kinetics - Mechanical, thermal and magnetic properties of solids	
Study Goals	The student: - Has an idea based on the atoms how a material is constructed, in terms of bonds and structure. - Can understand the construction of molecules and materials based on quantum chemical aspects. - Understands the basics regarding the presence of electronic bands in materials - Can indicate how diffusion of ions in solids is influenced. - Can discuss how thermal, mechanical and magnetic properties depend on the material composition. - Can discuss the factors that influence electrical and optical properties of semiconductors and metals. - Can make a selection of materials that are suitable for a specific application based on the above aspects	
Education Method	The course will consist of lectures and tutorials, in a 1-to-1 ratio. Exercises will be made and treated in the tutorials.	
Literature and Study Materials	Physics of Functional Materials by Hasse Fredriksson and Ulla Åkerlind, Wiley 2008. The following chapters will be discussed: 1-3 and 5-7.	
Assessment	Closed book exam at the end (written); formulary is provided.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

4052FYSTRY	Physical Transport Phenomena (FTV)	6
Responsible Instructor	Dr. B. Bera	
Instructor	Dr. V. Garbin	
Contact Hours / Week x/x/x/x	0/0/0/12	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The material of the lectures calculus (4051CALC1 and 4051CALC2), classical physics (from 4051QCHF), chemical thermodynamics (4051CHTHE), introduction to process technology (4051IPTEC) and is assumed to be known.	
Course Contents	Physical Transport phenomena forms the basis for many process technology courses. The course includes background and concepts with which physical/chemical processes and phenomena can be described/modeled.	
Study Goals	The college begins with the preservation laws of mass, energy and impulse. Be further dealt with: -Laws of Fick, Fourier and Newton -Dimension analysis; dimensionless Key Figures -Stationary and transient heat and mass transport (1-dimensional) -Heat transfer and mass transport over a phase boundary -Pressure drop calculation over a tube (system), laminar and turbulent flow The goal of the course is that the student: -knows the general recipe for the preparation of a mass, energy and impulse balance can apply to practical situations -is able to designate the transport mechanisms -is able to apply the substance of the college to issues of physical/chemical practice. -is able to make the right assumptions for modeling practical situation. -able to work models quantitatively. -is able to assess certain choice on their plausibility.	
Education Method	Compact lectures, e-learning (Open Course Ware TU Delft) environment in combination with the making of exercises.	
Literature and Study Materials	- H.E.A. van den Akker and R.F. Mudde, Transport Phenomena: The art of balancing; DUP (English version) - L.P.B.M. Janssen and M.M.C.G. Warmoeskerken, Transport Phenomena Data Companion, DUP	
Assessment	There are two ways to pass this course: Option 1 (Recommended): Option 1 is a combination of a group project and a final written exam. During the course, there will be a group assignment (4 or 5 persons per group). The group assignment will start from Week 3 and will finish in Week 8 of the quarter. This group assignment will cover the entire content of the course and will be divided in a number of stages. After every stage, the progress of the project will be evaluated and graded. Please note, there will not be a common group grade for this project. At each evaluation stage, the contribution of every member of a group will be discussed and graded accordingly. Hence, the grades of members of a group might very well be different. The grade of the group project will count for 40% of the final grade. At the end of the course, there will be a final written exam, covering the entire syllabus of the course. The grade of this written exam will count for 60% of the final grade. A student must obtain a minimum grade of 5.0 in the written exam. Option 2: It is possible not to take part in the group project. In that case, a student only takes part in the final written exam at the end of the course and 100% of the final grade comes from this written exam. A student must obtain a minimum grade of 5.5 to pass the final exam as well as the course.	
Permitted Materials during Tests	A simple calculator and the booklet "Transport Phenomena Data Companion" (without notes).	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	Central is acquiring the ability to translate a realistic problem into model comparisons and then to solve these comparisons. The required skill can only be acquired by practicing many example exercises. A few dozen are worked out in detail in the book. In addition, the bundle of issues offers practice assignments at (exam) level.	
Location	Delft	

4052LADIFY	Linear Algebra and Differential Equations	6
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	0/0/12/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	4051CALC1: Calculus 1 4051CALC2: Calculus 2	
Course Contents	First-order differential equations (emphasis on separable and linear differential equations), matrix algebra, Systems of linear equations, Gauss's elimination method, determinants, vectors in $\mathbb{R}^n$, linear transformations, eigenvalues, eigenspaces, matrix diagonalization, higher-order differential equations, systems of differential equations (emphasis on linear and autonomous systems), Laplace transformation, step functions, Dirac delta function, Fourier series, partial differential equations (emphasis on the heat equation, the wave equation and the Laplace equation)	
Study Goals	The student is able to: 1) perform the elementary matrix operations 2) solve a system of linear equations systematically 3) find the matrix representation of a linear transformation 4) determine the determinant, the eigenvalues and the eigenspaces of a square matrix and decide whether a square matrix is diagonalizable or not 5) solve linear differential equations using the Laplace transformation 6) solve systems of linear differential equations and visualize the typical solution curves 7) determine the Fourier series of a periodic function and solve simple partial differential equations	
Education Method	Colstructure	
Literature and Study Materials	- Linear Algebra, David Poole, fourth edition, - Elementary differential equations and boundary value problems, 11th edition (Global), Boyce en DiPrima - Dictaat (zie Brightspace)	
Assessment	During the course 2 summative partial tests (50% / 50%) will be taken that cover the entire content. In week 10 a resit will be offered for the entire content.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. PLEASE NOTE This is a course with a practical exercise: registration closes 28 days before the start of the first part of the course;	
Location	Delft	

4052THECHY	Theoretical Chemistry 1	6
Responsible Instructor	G.J. Kroes	
Instructor	Prof.dr. F. Buda	
Instructor	Dr. M. Somers	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Quantum Chemistry and Physics (QCF) and Calculus 1 + 2	
Course Contents	Elements of the mathematics of vector spaces, differential- and integral calculation, complex numbers. Introduction to the physical basis of quantum mechanics. Quantum mechanics (postulates and their interpretation, observables). The similarity between quantum mechanics and classical mechanics. Examples of quantum mechanical systems. The hydrogen atom. The variation theorem. The Independent Particle approach. Atoms with more electrons, electron spin, Pauli exclusion principle, Aufbau. The Born-Oppenheimer approach. Molecular orbital theory, secular equations. Molecular Orbitals-Linear Combination of Atomic Orbitals. The qualitative molecular orbital theory for homo- and hetero nuclear two atomic molecules. AH₂, AH₃ and AH₄ systems.	
Study Goals	After following Theoretical Chemistry, the student can: 1. apply the basic algebra that forms the basis for quantum mechanics, i.e., commutator algebra and simple linear algebra; 2. reproduce the postulates of quantum mechanics; 3. write down the equations (possibly to a proportionality constant) that describe how the energy of simple model systems depends on the quantum numbers that describe these systems; 4. explain, on the basis of simple model systems, the origin of specific quantum phenomena, such as zero-point energy and tunnelling; 5. describe what fermions and electron spin are, as well as the Pauli principle; 6. translate the Pauli principle into the mathematical form of the wave function of many-electron systems; 7. use simple mathematical techniques, such as separation of variables and application of orthogonality conditions, to derive relevant equations; 8. derive and explain in detail trends that are visible in the periodic table that are based on the Aufbau principle; 9. explain, based on the orbital approximation, what the Hartree method is and what the Hartree-Fock method is, and how those methods differ; 10. use the secular determinant and the secular equations to determine the energies of small molecules and the associated eigenfunctions; 11. use simple MO theory to determine the properties of diatomic molecules (for example, whether A₂ molecules are paramagnetic or not) and of triatomic molecules (for example, the shape of AH₂, where A = Li-F); 12. explain what the LCAO method is; 13. explain how the Born-Oppenheimer approach can be used for theoretical descriptions of molecular spectroscopy and the dynamics of chemical reactions.	
Education Method	Classroom lectures, tutorials and computer software	
Literature and Study Materials	- Colledictaat (download possible on Brightspace) - Physical Chemistry van T. Engel en P. Reid (Pearson: Benjamin Cummings)	
Assessment	The result for TC1 is determined as follows. Based on 2 submitted tutorial sums (werkcollege opgaven) becomes 5% of the grade certain. 5% of the mark is determined on the basis of the 2 computer practicals. 90% of the grade is determined on the basis of the examination. For a sufficient is a final grade of at least 5.5 required.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Major Synthese BSc MST 2e jaar 2022

4052BMOCHY	Biomolecular Chemistry (BMC)	6
Responsible Instructor	Dr. J.D.C. Codee	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	4051OCSTRY: Organische Chemie en Structuuranalyse 4052ORGC2Y: Organische Chemie 2 (OC 2)	
Course Contents	During this course a number of important biological molecules will be discussed from a (organic) chemical perspective. The biosynthesis of amino acids, sugars, nucleic acids and lipids and their role in the biosynthesis and degradation of complex biopolymers will be discussed. Certain complex biological processes, such as phosphorylating paths and post-translational modifications will come to order. A broad frame of view will be made for each biological process. Zooming in, a set of specific chemical processes and their mechanisms behind it will be discussed.	
Education Method	lecture and tutorial	
Literature and Study Materials	Lecture notes downloadable from Brightspace: Biomolecular Chemistry - Leeuwenburg, Overkleef, Codée	
Assessment	Written exam and retake.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

4052FYSC6	Physiological Chemistry	6
Responsible Instructor	Prof.dr. J.M.F.G. Aerts	
Instructor	Dr. R.R.C.L. Olsthoorn	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Biochemistry	
Course Contents	The course physiological chemistry is concerned with biochemical processes in relation to diseases. The student will be taught the molecular and cellular basis of acquired and inheritable anomalies in lipids, proteins, sugars and nucleic acids.	
Study Goals	1. Knowledge The student is familiar with the basic terms of physiological chemistry. 2. Insight and understanding The student will be given insight into the cellular molecular processes engaged in: protein synthesis, control and degradation, metabolism of lipids and sugars, hormonal regulation of metabolism and its connection to the brain, by using concrete examples. 3. Application and analysis The student is capable of applying the taught material on questions and cases from physiological chemistry.	
Education Method	- Lectures and presentation / symposium	
Literature and Study Materials	- Lehninger Principles of Biochemistry, Nelson and Cox, 7th edition (2017), ISBN-13: 978-1-4641-2611-6, Macmillan. - Brightspace. - Recommended: Molecular Biology of the Cell, 6th edition, Alberts, Johnson, Lewis, Morgan, Raff, Roberts, Walter, Garland Science, ISBN 978-0-8153-4464-3	
Assessment	2 partial exams that each account for 40% of the final mark for which at least a 5.5 must be passed. Oral presentation for fellow students counts for 20%. The resit exam covers the entire material.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

4052ORGC2Y	Organic Chemistry 2 (OC 2)	6
Responsible Instructor	M. Overhand	
Instructor	Dr. S. van der Vorm	
Instructor	Dr. R.J.B.H.N. van den Berg	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	This lecture series is a follow up of the first year course organic chemistry 1. A number of new reactions are introduced, such that the student obtains a complete picture of the fundamental reactions and reaction mechanisms of organic chemistry: - nucleophilic (aromatic) substitution; - electrophilic aromatic substitution; - elimination reactions; - nucleophilic- and electrophilic addition; - cyclo addition; - radical reactions. Both the conditions and the mechanisms of reactions, prediction of the reaction product and retro-synthetic analysis are treated in this course. The specific subjects treated are: - Diels-Alder addition; - Wittig reaction; - nitration; - diazotations; - Sandmeyer reaction; - reductive amination; - amino acids; - pI calculation; - enzyme mechanisms, role of cofactors; - Friedel-Crafts alkylation and acylation; - alkylation of enolates; - Dieckmann rearrangement; - hard and soft nucleophiles.	
Study Goals	Upon completion of the course, the student: - is familiar with the general concepts of organic chemistry, such as: nucleophile, electrophile, hard and soft, leaving groups, pKa, intramolecular, cyclo-addition; - possesses a working knowledge of the reactivity of a broad range of organic compounds; - has a clear understanding of the relative reactivity of functional groups; - understands the underlying mechanisms of organic reactions; - is able to perceive relations with other chemical sub-disciplines.	
Education Method	Lecture and tutorial	
Literature and Study Materials	Organic Chemistry, Clayden, Greeves, Warren and Wothers, Oxford press 2001	
Assessment	Written exam and retake. LET OP: In verband met de maatregelen rondom het Coronavirus kan gedurende het studiejaar een wijziging optreden in de wijze van toetsing. LET OP: indien er wederom maatregelen nodig zijn in verband met COVID, dan worden de werkcolleges verplicht en dan kunnen de studenten de antwoorden op de werkcolleges inleveren als huiswerk voor bonuspunten.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

4052THECHY	Theoretical Chemistry 1	6
Responsible Instructor	G.J. Kroes	
Instructor	Prof.dr. F. Buda	
Instructor	Dr. M. Somers	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	Quantum Chemistry and Physics (QCF) and Calculus 1 + 2	
Course Contents	Elements of the mathematics of vector spaces, differential- and integral calculation, complex numbers. Introduction to the physical basis of quantum mechanics. Quantum mechanics (postulates and their interpretation, observables). The similarity between quantum mechanics and classical mechanics. Examples of quantum mechanical systems. The hydrogen atom. The variation theorem. The Independent Particle approach. Atoms with more electrons, electron spin, Pauli exclusion principle, Aufbau. The Born-Oppenheimer approach. Molecular orbital theory, secular equations. Molecular Orbitals-Linear Combination of Atomic Orbitals. The qualitative molecular orbital theory for homo- and hetero nuclear two atomic molecules. AH₂, AH₃ and AH₄ systems.	
Study Goals	After following Theoretical Chemistry, the student can: 1. apply the basic algebra that forms the basis for quantum mechanics, i.e., commutator algebra and simple linear algebra; 2. reproduce the postulates of quantum mechanics; 3. write down the equations (possibly to a proportionality constant) that describe how the energy of simple model systems depends on the quantum numbers that describe these systems; 4. explain, on the basis of simple model systems, the origin of specific quantum phenomena, such as zero-point energy and tunnelling; 5. describe what fermions and electron spin are, as well as the Pauli principle; 6. translate the Pauli principle into the mathematical form of the wave function of many-electron systems; 7. use simple mathematical techniques, such as separation of variables and application of orthogonality conditions, to derive relevant equations; 8. derive and explain in detail trends that are visible in the periodic table that are based on the Aufbau principle; 9. explain, based on the orbital approximation, what the Hartree method is and what the Hartree-Fock method is, and how those methods differ; 10. use the secular determinant and the secular equations to determine the energies of small molecules and the associated eigenfunctions; 11. use simple MO theory to determine the properties of diatomic molecules (for example, whether A₂ molecules are paramagnetic or not) and of triatomic molecules (for example, the shape of AH₂, where A = Li-F); 12. explain what the LCAO method is; 13. explain how the Born-Oppenheimer approach can be used for theoretical descriptions of molecular spectroscopy and the dynamics of chemical reactions.	
Education Method	Classroom lectures, tutorials and computer software	
Literature and Study Materials	- Collegedictaat (download possible on Brightspace) - Physical Chemistry van T. Engel en P. Reid (Pearson: Benjamin Cummings)	
Assessment	The result for TC1 is determined as follows. Based on 2 submitted tutorial sums (werkcollege opgaven) becomes 5% of the grade certain. 5% of the mark is determined on the basis of the 2 computer practicals. 90% of the grade is determined on the basis of the examination. For a sufficient is a final grade of at least 5.5 required.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Leiden	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Major Technologie BSc MST 2e jaar 2022

4052CHBIOY	Chemical Biotechnology (CBT)	6
Responsible Instructor	Dr. L. Jourdin	
Instructor	Dr.ir. A.J.J. Straathof	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Required for	Biotechnology-related minors or BSc thesis projects.	
Expected prior knowledge	Chemische Thermodynamica, Inleiding Procestechnologie, Biochemie, Fysische Transportverschijnselen, Fysische chemie en Kinetiek	
Course Contents	Introduction on industrial and environmental biotechnology; use of enzymes, pure cultures, mixed cultures; importance of biomass and sustainability Nutrients needed for growth of microorganisms Stoichiometry of microbial conversions according to mass balances Hyperbolic kinetics for substrate uptake Herbert-Pirt distribution of substrate between growth, product formation and maintenance; relations between cell mass-specific rates (μ and q) Black-box model for aerobic and anaerobic product formation Determination of the black-box model parameters from batch and chemostat experiments The fermentor, and determination of relevant transport rates Design of fermentation in batch and chemostat Energy conversion pathways Deriving the product pathway reaction Deriving the biomass reaction Microbial competition; application in mixed cultures Designing enzymatic conversions	
Study Goals	- The student will be able to quantitatively analyze and evaluate the most important requirements for microbial growth and product formation. - The student will be able to apply simple biokinetic, stoichiometric, and mass/heat transport relationships in calculations with ideal reactors. - The student will be able to derive process reactions from data on growth reaction, product pathway reaction, and energy conversion pathways.	
Education Method	Relevant parts of the MOOC (massive open online courseware) Industrial Biotechnology Classroom excercises	
Literature and Study Materials	Reader Online courseware Old exam questions	
Assessment	Assignments (15% final grade) and exam (85%). Resit.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

4052FYSTRY	Physical Transport Phenomena (FTV)	6
Responsible Instructor	Dr. B. Bera	
Instructor	Dr. V. Garbin	
Contact Hours / Week x/x/x/x	0/0/0/12	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The material of the lectures calculus (4051CALC1 and 4051CALC2), classical physics (from 4051QCHF), chemical thermodynamics (4051CHTHE), introduction to process technology (4051IPTEC) and is assumed to be known.	
Course Contents	Physical Transport phenomena forms the basis for many process technology courses. The course includes background and concepts with which physical/chemical processes and phenomena can be described/modeled.	
Study Goals	The college begins with the preservation laws of mass, energy and impulse. Be further dealt with: -Laws of Fick, Fourier and Newton -Dimension analysis; dimensionless Key Figures -Stationary and transient heat and mass transport (1-dimensional) -Heat transfer and mass transport over a phase boundary -Pressure drop calculation over a tube (system), laminar and turbulent flow The goal of the course is that the student: -knows the general recipe for the preparation of a mass, energy and impulse balance can apply to practical situations -is able to designate the transport mechanisms -is able to apply the substance of the college to issues of physical/chemical practice. -is able to make the right assumptions for modeling practical situation. -able to work models quantitatively. -is able to assess certain choice on their plausibility.	
Education Method	Compact lectures, e-learning (Open Course Ware TU Delft) environment in combination with the making of exercises.	
Literature and Study Materials	- H.E.A. van den Akker and R.F. Mudde, Transport Phenomena: The art of balancing; DUP (English version) - L.P.B.M. Janssen and M.M.C.G. Warmoeskerken, Transport Phenomena Data Companion, DUP	
Assessment	There are two ways to pass this course: Option 1 (Recommended): Option 1 is a combination of a group project and a final written exam. During the course, there will be a group assignment (4 or 5 persons per group). The group assignment will start from Week 3 and will finish in Week 8 of the quarter. This group assignment will cover the entire content of the course and will be divided in a number of stages. After every stage, the progress of the project will be evaluated and graded. Please note, there will not be a common group grade for this project. At each evaluation stage, the contribution of every member of a group will be discussed and graded accordingly. Hence, the grades of members of a group might very well be different. The grade of the group project will count for 40% of the final grade. At the end of the course, there will be a final written exam, covering the entire syllabus of the course. The grade of this written exam will count for 60% of the final grade. A student must obtain a minimum grade of 5.0 in the written exam. Option 2: It is possible not to take part in the group project. In that case, a student only takes part in the final written exam at the end of the course and 100% of the final grade comes from this written exam. A student must obtain a minimum grade of 5.5 to pass the final exam as well as the course.	
Permitted Materials during Tests	A simple calculator and the booklet "Transport Phenomena Data Companion" (without notes).	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	Central is acquiring the ability to translate a realistic problem into model comparisons and then to solve these comparisons. The required skill can only be acquired by practicing many example exercises. A few dozen are worked out in detail in the book. In addition, the bundle of issues offers practice assignments at (exam) level.	
Location	Delft	

4052LADIFY	Linear Algebra and Differential Equations	6
Responsible Instructor	Dr. B.J. Meulenbroek	
Contact Hours / Week x/x/x/x	0/0/12/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	4051CALC1: Calculus 1 4051CALC2: Calculus 2	
Course Contents	First-order differential equations (emphasis on separable and linear differential equations), matrix algebra, Systems of linear equations, Gauss's elimination method, determinants, vectors in $\mathbb{R}^n$, linear transformations, eigenvalues, eigenspaces, matrix diagonalization, higher-order differential equations, systems of differential equations (emphasis on linear and autonomous systems), Laplace transformation, step functions, Dirac delta function, Fourier series, partial differential equations (emphasis on the heat equation, the wave equation and the Laplace equation)	
Study Goals	The student is able to: 1) perform the elementary matrix operations 2) solve a system of linear equations systematically 3) find the matrix representation of a linear transformation 4) determine the determinant, the eigenvalues and the eigenspaces of a square matrix and decide whether a square matrix is diagonalizable or not 5) solve linear differential equations using the Laplace transformation 6) solve systems of linear differential equations and visualize the typical solution curves 7) determine the Fourier series of a periodic function and solve simple partial differential equations	
Education Method	Colstructure	
Literature and Study Materials	- Linear Algebra, David Poole, fourth edition, - Elementary differential equations and boundary value problems, 11th edition (Global), Boyce en DiPrima - Dictaat (zie Brightspace)	
Assessment	During the course 2 summative partial tests (50% / 50%) will be taken that cover the entire content. In week 10 a resit will be offered for the entire content.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. PLEASE NOTE This is a course with a practical exercise: registration closes 28 days before the start of the first part of the course;	
Location	Delft	

4052SCHTEY	Separation Technology	6
Responsible Instructor	Dr.ir. M.A. van der Veen	
Instructor	D.A. Vermaas	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3 4	
Exam Period	4	
Course Language	Dutch	
Expected prior knowledge	The teachers expect that the material of the first-year colleges of calculus, chemical thermodynamics and introduction to process technology is known and that the student has this ready. A number of aspects that are essential to the course are used immediately in the first test. The results of the first test are therefore seen as a suitability criterion for further participation in the course.	
Course Contents	The Separation Technology course can be seen as the core of Chemistry: after each synthesis, one or more processes necessarily follow to separate the product from by-products and residual reactants, solvents and other additives. This course deals with the six most important separation processes, of which the first five are based on phase equilibria. 1. Distillation 2. Extraction 3. Absorption 4. Adsorption 5. Crystallization 6. Membrane separations Because thermodynamic phase equilibrium ultimately indicates which level of separation is aimed for by a system and moreover indicates how much work is required as a minimum, the following thermodynamic topics precede the relevant separation processes. And also, because modelling of chemical processes, requires a correct treatment of their thermodynamics, much attention is devoted to the fundamentals of the thermodynamics of following systems. 1. Phase equilibrium 2. Ideal and non-ideal liquid vapor equilibrium 3. Non-ideal liquid-liquid equilibrium 4. Crystallization, liquid-solid equilibrium	
Study Goals	1. Knowledge a. The student is familiar with the most important industrial separation processes and the thermodynamics of physical and chemical equilibrium. b. The student is familiar with the general recipe for drawing up mass and energy balances. 2. Insight and understanding a. The student is able to interpret phase diagrams for ideal and non-ideal, reactive and non-reactive binary and ternary systems based on thermodynamic information available in the literature and to prepare them where necessary. The student is capable to describe aforementioned systems thermodynamically. b. The student is able to make a selection based on knowledge of basic principles. c. The student is able to model the various forms of transport. 3. Application and analysis of the substance a. The student is able to quantify the quality and yield of a separation process on the basis of the phase information provided. b. The student is able to make the correct assumptions for modeling practical situations. c. The student is able to quantify models from works, i.e. developing phase diagrams based on thermodynamic models and data, calculating mass and energy flows, to be able to determine basic dimensions of separation devices. 4. Synthesis and evaluation a. The student is able to apply the basic principles to new situations and to assess the plausibility of a certain choice on its plausibility. b. The student is able to prepare and assess the analysis of a (part of a) separation process and to i. Method: the application of available theoretical or derived information; ii. Numerical elaboration with the help of literature or experimental data; iii. Plausibility of the accuracy and order of magnitude of the answers obtained.	
Education Method	Compact lectures in combination with tutorials and guided self-study. Tutorial videos on the basic concepts will be provided, which the student is expected to study before the lecture. The lecture will mainly consist of discussion and application of the material. During the supervised self-study (tutorials) assignments are handed out that can / must be made on the spot. In addition, a number of assistants are available for answering questions and giving hints and explanations.	
Literature and Study Materials	- André B. de Haan, Burak Eral, Boelo Schuur, Industrial Separation Processes: Fundamentals, De Gruyter, 2020. - M. Scott Shell, Thermodynamics and Statistical Mechanics, Cambridge University Press, 2015.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	Central is acquiring the ability to translate a realistic problem into a model and then to solve the resulting comparisons. The required skill can only be acquired by practicing (very) many example assignments. In the tutorials, a few dozen are worked out in detail and some will be discussed during the lectures. Analogous, it is necessary to acquire the skill to describe different systems correctly thermodynamically. This skill can only be acquired via describing many different examples correctly, and to each time understand explicitly why the theory is applied in that manner. Several dozens of these examples are worked out in the exercise sessions.	
Location	Delft	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

3e jaar BSc MST 2022

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Kernprogramma BSc MST 3e jaar 2022

4052LEON3	Chemical Product Design	6
Responsible Instructor	Dr.ir. G.M.H. Meesters	
Instructor	Dr. S. Saedy	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Course Contents	The Chemical Product Design course provides insight into the aspects that play a role in the development of chemical products. Teams of 5-6 students work on designing a product. The goal is to effectively use the creativity and acquired knowledge about chemical engineering of the team to develop an innovative but realistic new product. The following topics are covered in the course: -Working efficiently in a team -Aspects that are important for product design -The relationship between product properties and physico-chemical composition -Development of physico-chemical models to describe product properties. -Identifying customer needs -Generating new ideas. -Selecting the most promising idea. -Developing a new product. The course mimics the situation in a chemical company, where a new product is developed. Your team is a group of engineers who have received this task from their manager. The role of the manager is played by a staff member of the Chemical Engineering Department (the mentor).	
Study Goals	- General Skills A. The student can work (creatively) in a team. B. The student is able to report the results of the (team) work in writing, to present and to discuss orally. - Knowledge A. The student is familiar with the basic concepts of product design. - Understanding A. The student is able to indicate the relationship between physical-chemical properties and product properties. B. The student is able to identify a new product. - Application and analysis of the content A. The student is able to apply the content of the lecture in the field of chemical process/product technology. B. The student is able to make a quantitative physico-chemical model of relevant product characteristics. - Synthesis and evaluation A. Using the knowledge and skills acquired in this course the student is able to design new products.	
Education Method	Lectures and a group project (working in a team).	
Literature and Study Materials	Design & Development of Biological, Chemical, Food and Pharmaceutical Products, Wesselingh and Kiil, 2007, John Wiley and Sons Ltd, ISBN 9780470061558	
Prerequisites	Deelname aan Leren Onderzoeken 3, het Chemisch Product Ontwerp (CPO), is pas mogelijk na succesvol afronden van alle vakken van het eerste jaar en ten minste 30 EC uit het tweede studiejaar.	
Assessment	The course is tested by means of an assignment about working in a team, a written report about the functioning of an existing product (P1). A written report about the design of your own product and a final presentation about this own product design (P2). The weighting of P1 and P2 to the final result is as follows: - P1 counts for 20%, P2 for 80%. - P1 is a report. - The P2 result is 75% determined by a report and 25% by a presentation. - All group grades are made individually via a peer review.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. PLEASE NOTE This is a course with a practical exercise: registration closes 28 days before the start of the period.	

4052STAMEY	Statistical Methods (STM)	3
Responsible Instructor	Dr. C.O. Smet	
Instructor	L. Vuursteen	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Expected prior knowledge	4051CALC1Y: Calculus 1 4051CALC2Y: Calculus 2	
Course Contents	Graphical and numerical summaries of data Introduction to probability Some elementary combinatorics to find probabilities Conditional probability; Bayes' rule; stochastic (in)dependence Random variables; probability mass function; density function; distribution function Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto Expectation and variance; transformations; Jensen's inequality Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables Covariance and correlation Chebychev's inequality; Law of Large numbers; Central Limit Theorem Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot Theory of Estimators: Bias, Efficiency, Mean squared error Linear Regression: univariate and multivariate, goodness of fit Introduction to statistical learning Confidence intervals of Mean and Proportion; t-distribution Testing theory: Type I/II Error, p-value, significance level, critical region One sample t-test	
Study Goals	At the end of the course, the students will be able to: - compute (conditional) probabilities - use discrete and continuous random variables - use joint random variables - compute expectation and (co)variance - apply the law of large numbers and the central limit theorem - analyse data by means of graphical and numerical summaries - formulate a statistical model - estimate parameters - perform a linear regression analysis - construct a confidence interval - formulate and carry out hypothesis tests - Use Python to perform elementary simulation and data analysis	
Education Method	Students watch last year's recordings and make the exercises. There is room for questions and additional exercises during one 2-hour lecture each week.	
Literature and Study Materials	A Modern Introduction to Probability and Statistics Understanding Why and How Series: Springer Texts in Statistics Dekking, F.M., Kraaikamp, C., Lopuhaa, H.P., Meester, L.E. 1st ed. 2005. Corr. 2nd printing, 2007. ISBN: 978-1-85233-896-1	
Assessment	One final test at the end of Q1, one resit opportunity at the end of Q2.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Location	Delft	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Major Synthese en Materialen BSc MST 3e jaar 2022

4052LEON4	Bachelor Thesis Project	15
Module Manager	M.P. Swarte	
Contact Hours / Week x/x/x/x	0/0/0/40	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch English	
Course Contents	Students work individually on a project in the context of ongoing research at one of the research groups.	
Study Goals	1. Literature Review/study: Is able to study relevant scientific literature and to look up data sources and assess them by value. 2. Preparing/Setting up research: -Is able to prepare experiments (formulation research question, setting up research plan) -Environment and safety: can work safely with chemicals, taking into account their physical and chemical properties, including any specific hazards associated with the use of these materials -Is skilled in making a risk analysis of the use of chemicals 3. Collecting/Generating Research data: -Is able to perform standard laboratory prescriptions -May carry out observations and measurements under limited supervision, as well as their correct reporting (including a well-kept lab journal) 4. Analysing and processing of research material: -He/she can interpret the observations and data obtained in the laboratory and relate it to existing and/or an appropriate theory -Ethics: Has an understanding of his/her professional and ethical responsibility 5. Working posture and teamwork -Dedication, motivation, enthusiasm, creativity -Able to function in a (multicultural and multidisciplinary) team -Is able to actively participate, cooperate effectively, effectively discuss. 6. Reporting: -Able to report in writing and orally with attention to requirements for scientific communication and accountability -Able to communicate effectively in Dutch and English, using modern presentation techniques	
Education Method	Practical work in a research group including attending work discussions, presenting own results, self-study and teamwork.	
Prerequisites	You can only start your final project if the registration form has been approved by the study advisor and the responsible staff member or examiner. The form will only be approved if you meet the following admission requirements: - First year completed. - LO2 achieved. - Achieved at least 18 points theory of specialization Materials, Synthesis or Technology. - Achieved at least 24 points of the 2nd year compulsory program.	
Assessment	The assessment takes place on the basis of the learning objectives. A rubric is used for this (ask this at the start of your final project). The final paper is presented in writing (report) and orally (presentation) to a committee of experts (one from outside the section). After the report has been defended, the final assessment is determined and the student receives an explanation of how the assessment was made.	

4052STTS6	Statistical Thermodynamics / Spectroscopy	6
Responsible Instructor	Dr. E.M. Blokhuis	
Instructor	Dr. D.G.H. Hetterscheid	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	Dutch	
Expected prior knowledge	.	
Course Contents	The course Statistical Thermodynamics and Spectroscopy comprises two parts: - Statistical Thermodynamics (Edgar Blokhuis) - Spectroscopy (Dennis Hetterscheid) * As the name suggests, the first part on the Statistical Thermodynamics very much builds on the knowledge acquired in an introductory course in Thermodynamics, but now we delve deeper in the underlying world of atoms and molecules and see that all thermodynamics quantities, such as temperature or free energy, carry microscopic interpretation in terms of the movement of molecules and atoms. Chapter 1: a short review of Thermodynamics Chapter 2: the link between Thermodynamics and Statistical Thermodynamics is discussed leading to a formalism that allows us to calculate the free energy from all the available states of the system. Chapter 3: The formalism derived in Chapter 3 is applied to the Ideal Gas. It is shown how Statistical Thermodynamics provides a way to calculate equilibrium constants for chemical reactions from spectroscopic data. * Spectroscopy is a collective term for scientific techniques that investigate the interaction of matter with irradiation. In this part of the course the general principles of spectroscopy are explained and the student will learn to correlate spectroscopic data with the quantum mechanical description of the matter investigated, and vice versa. Main topics are: molecular symmetry, group theory, atomic spectroscopy, UV/Vis spectroscopy, infrared spectroscopy and Raman spectroscopy.	
Study Goals	Learning objectives for spectroscopy: - The student is able to determine the (most important) quantum states of a given material (atoms, small molecules) and can assign these states to energy Terms. - The student is able to determine which quantum state(s) belong(s) to the ground state. - The student can rationalize which transitions between quantum states as a result of an absorption, emission or scattering event have a more than zero probability of taking place. - The student can determine which atomic orbitals, molecular orbitals, vibrational normal modes and energy terms belong to a particular irreducible representation and rationalize from there which transitions in vibrational and electronic spectroscopy are symmetry forbidden. - The student is able to determine which vibrational motion belongs to a particular transition in the infrared region. - The student is able to qualitatively predict which signals are to be observed in the rotational, vibrational or electronic spectrum of various materials ranging from single atoms (atomic spectroscopy) to large molecules (IR, Raman, UV-vis spectroscopy).	
Education Method	Lectures and tutorials	
Literature and Study Materials	* Thomas Engel and Philip Reid, Physical Chemistry, Pearson, 3rd Edition, 978-1-292-0222406 * Alan Vincent, Molecular Symmetry and Group Theory, 2nd Edition, 978-0-471-48939-9 * Slides presented during the courses * Exercises	
Assessment	Both parts end with a written partial exam. The grades for both partial exams are determined to one decimal place. For one of the two partial exams a grade between 5.0-5.4 is allowed when it is compensated by the grade of the other partial exam. The final grade is determined by the average of the two (partial) exams and rounded to half grade points. Each partial exam counts for 50%. The retake exam concerns the content of the *whole* course. The grades of the partial exams are valid only in the current study year.	
Enrolment / Application	Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	

Year	2022/2023
Organization	Applied Sciences
Education	Bachelor Molecular Science and Technology

Major Technologie BSc MST 3e jaar 2022

4052CHREKY	Chemical Reactor Engineering	6
Responsible Instructor	Prof.dr. A. Urakawa	
Instructor	Dr. T.E. Burdyny	
Contact Hours / Week x/x/x/x	0/0/8/0	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	IPT, Calculus 1 and 2, Physical Chemistry and Kinetics, Physical Transport Phenomena	
Course Contents	- Design equations for ideal reactors - Conversion and selectivity in ideal reactors - Reactors in series and recycle reactor - Relation among kinetics, reaction networks and reactor performance - Heat effects: Adiabatic reactor operation - Non-ideal reactors and residence time distribution - Basics of catalytic reactions	
Study Goals	The student is able to: - Apply mass and energy balances to chemical reactors - Have a notion of the characteristics of basic reactor types - Predict/calculate the conversion and selectivity - Design & select a chemical reactor - Predict optimal reaction conditions - Extract the kinetics of the reactions from experiments - Use and apply numerical techniques, next to back of the envelope calculations	
Education Method	Lectures, tutorials and homework	
Computer Use	Some complex problems require a use of computer and programming skills to solve them. Python, MATLAB and Excel can be used for the purpose among others, with a recommended use of Python. Computer-based problems will be given for homework but not for exams.	
Course Relations	This course builds upon IPT, and the student is expected to be quick and proficient in setting up material and energy balances. Some of the topics of this course are also treated in Transport Phenomena. The focus here is on integrating these ideas and knowledge for reactor design, which makes this a math-oriented subject. Skills to solve differential equations are assumed. After this course, students will be ready for the next step; seeing how design aspects for reactors integrate with design aspects of a variety of products and processes.	
Literature and Study Materials	Octave Levenspiel: Chemical Reaction Engineering, 3rd edition Lecture notes (Brightspace)	
Assessment	Mid-term exam (30%), final exam (55%) and homework (15%) Resit will account for 100% of the score (homework and mid-term exam scores will not be taken into account)	
Enrolment / Application	ATTENTION: Due to measures taken in the framework of the Coronavirus the way of examination could differ Registration in uSis is required to participate in this course, use My StudyMap. The standard registration period of 14 calendar days before the first activity of the course applies to this course. PLEASE NOTE: registration for (re)examinations only concerns a pre-registration, your (re)examination registration is only final if you have confirmed it via My StudyMap. NB Only register for the components that you are actually going to follow, it is also mandatory to unsubscribe if you do not participate.	
Remarks	English is the language used in the course and in the exam	
Studyload/Week	The course is designed to comply with a study load of 28 hours per ECTS for a student of average talent. We structure the course contents with a good amount of prescribed homework to practice, which is corrected and marked by the teaching staffs and assistants. The homework is the key addition to the lectures and self-study.	
Location	Delft	

4052LEON4	Bachelor Thesis Project	15
Module Manager	M.P. Swarte	
Contact Hours / Week x/x/x/x	0/0/0/40	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch English	
Course Contents	Students work individually on a project in the context of ongoing research at one of the research groups.	
Study Goals	1. Literature Review/study: Is able to study relevant scientific literature and to look up data sources and assess them by value. 2. Preparing/Setting up research: -Is able to prepare experiments (formulation research question, setting up research plan) -Environment and safety: can work safely with chemicals, taking into account their physical and chemical properties, including any specific hazards associated with the use of these materials -Is skilled in making a risk analysis of the use of chemicals 3. Collecting/Generating Research data: -Is able to perform standard laboratory prescriptions -May carry out observations and measurements under limited supervision, as well as their correct reporting (including a well-kept lab journal) 4. Analysing and processing of research material: -He/she can interpret the observations and data obtained in the laboratory and relate it to existing and/or an appropriate theory -Ethics: Has an understanding of his/her professional and ethical responsibility 5. Working posture and teamwork -Dedication, motivation, enthusiasm, creativity -Able to function in a (multicultural and multidisciplinary) team -Is able to actively participate, cooperate effectively, effectively discuss. 6. Reporting: -Able to report in writing and orally with attention to requirements for scientific communication and accountability -Able to communicate effectively in Dutch and English, using modern presentation techniques	
Education Method	Practical work in a research group including attending work discussions, presenting own results, self-study and teamwork.	
Prerequisites	You can only start your final project if the registration form has been approved by the study advisor and the responsible staff member or examiner. The form will only be approved if you meet the following admission requirements: - First year completed. - LO2 achieved. - Achieved at least 18 points theory of specialization Materials, Synthesis or Technology. - Achieved at least 24 points of the 2nd year compulsory program.	
Assessment	The assessment takes place on the basis of the learning objectives. A rubric is used for this (ask this at the start of your final project). The final paper is presented in writing (report) and orally (presentation) to a committee of experts (one from outside the section). After the report has been defended, the final assessment is determined and the student receives an explanation of how the assessment was made.	

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